Docker Networking - Practical Lab

Luma University | Learning & Grow | Synthetic demo material

Learning outcomes

Create a reproducible API and PostgreSQL environment with Docker Compose.
Explain service-name resolution within a Compose network.
Troubleshoot readiness and keep credentials out of source control.

Lab brief

Build a FastAPI service with a GET /health endpoint. Define API and database services in compose.yaml. Services in the same Docker Compose network can address each other by service name; localhost inside a container refers to that container itself.

Network configuration

Use db as the database hostname for the API connection. Publish only the API port to the host. Keep the database port private and provide credentials through environment variables. Add a database health check before relying on its readiness.

Verification

Run docker compose up --build and inspect service logs. Confirm the API can connect to PostgreSQL and return a healthy response. Stop and recreate the containers to verify the setup is repeatable.

Deliverables

Submit compose.yaml, a sanitized configuration example and a README containing startup, test and troubleshooting commands. Include a brief explanation of why service-name resolution works. Allow 45 minutes for the lab and 15 minutes for the README.

Fictional learning material and credentials for demonstration.